Machine Learning Engineer Job Description: Top Duties and Qualifications

Edited by Indeed Employer Content Team

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On this page

- Machine Learning Engineer duties and responsibilities
- Machine Learning Engineer skills and qualifications
- Machine Learning Engineer experience requirements
- Machine Learning Engineer education and training requirements

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Machine Learning Engineer duties and responsibilities

The primary task of a Machine Learning Engineer is to create AI capable of mimicking human thought processes as closely as possible so that they can function reliably and effectively. Doing so involves the following Machine Learning Engineer job duties:

- Working with project managers to establish objectives for AI systems
- Creating AI algorithm prototypes based on project specifications
- Running tests to assess AI performance
- Analyzing data gathered during tests to identify strengths and weaknesses of AI
- Implementing changes to algorithms to improve AI performance
- Troubleshooting and addressing problems with deployed AI to improve user experience
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Machine Learning Engineer skills and qualifications

To effectively carry out Machine Learning Engineer job duties, candidates need certain skills. Some qualifications that are common among quality candidates include:

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- Rigorous knowledge of R, Java and Python programming to develop software
- In-depth knowledge of machine learning data structures and modeling, software architecture, libraries and frameworks to create AI that accomplishes outlined goals
- Strong mathematics skills, especially in statistics, to create algorithms
- Effective oral and written communication skills to collaborate with others and write detailed, easy-to-read reports
- Advanced analytical skills to interpret test results and performance metrics
- Creative thinking skills to come up with new solutions and approaches
- Sound problem-solving skills to refine prototypes and troubleshoot performance issues
- Ability to remain stationary for long periods of time in front of screens

Machine Learning Engineer experience requirements

How much experience you should look for in candidates for your Machine Learning Engineer position depends on your company's needs. If the candidate will work with a team of other Engineers who have many years of experience, they may only need one to two years of related experience in programming or machine learning. New hires who'll work independently or join a team with other new Engineers may need two to five years of experience working specifically in machine learning to perform their job duties effectively.

Machine Learning Engineer education and training requirements

A bachelor's degree in artificial intelligence, machine learning, computer science, data science, statistics or mathematics is usually necessary to work in Machine Learning Engineer careers. You may wish to prioritize candidates who have more education, such as a master's degree or post-graduate certificate. Individuals who hold AWS Certified Machine Learning certification from Amazon or Professional Machine Learning Engineer certification from Google have passed written examinations, demonstrating that they likely possess the knowledge required to succeed as a Machine Learning Engineer.

Machine Learning Engineer salary expectations

According to Indeed Salaries, the average salary for a Machine Learning Engineer is $131,076 per year. The actual annual income of an Engineer may be more or less based on geographic location, experience, education and certifications.

Job description samples for similar positions

If the Machine Learning Engineer position doesn't match the requirements of your company, read the duties and responsibilities for these related jobs:

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- Computer Engineer
- IT Business Analyst
- IT Project Manager

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Machine Learning Engineer job description FAQs

Who does a Machine Learning Engineer report to?

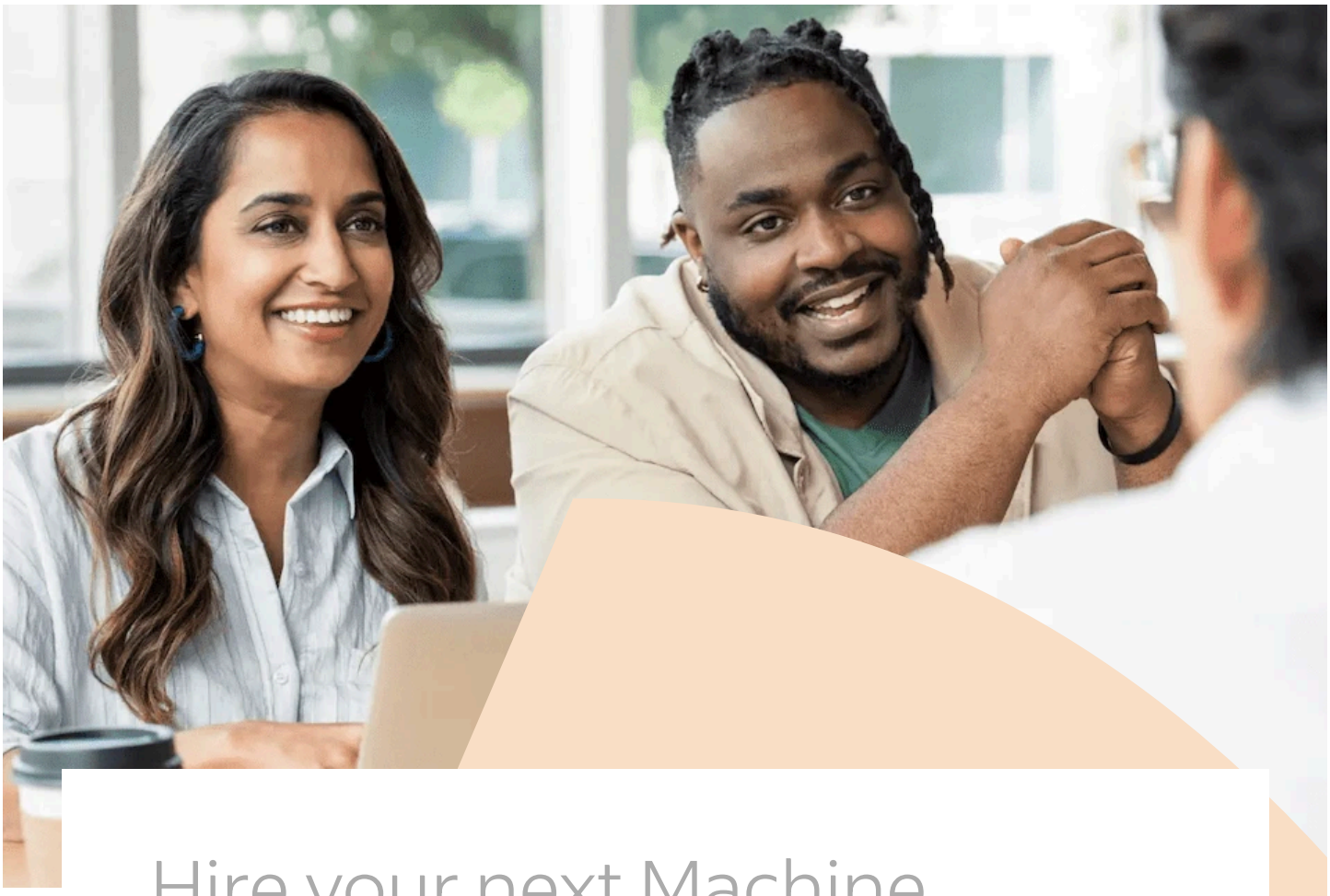
A Machine Learning Engineer usually reports to whoever is overseeing AI development at a company. Depending on organizational structures, machine learning may be in its own department, or it may fall under the category of research and development or information technology (IT). The supervisor for a Machine Learning Engineer may have the title of Senior Machine Learning Engineer, Machine Learning Manager or Manager of Research and Development. In some cases, a Machine Learning Engineer may answer directly to a project manager.

How can you make your Machine Learning Engineer job description stand out?

Due to the rise in SMART technology and the growth of ecommerce, Machine Learning Engineers are in demand. To make your job description stand out and encourage quality candidates to apply for a Machine Learning Engineer role, work to attract your ideal candidate. People who perform well under pressure and can make quick, decisive actions are often well-suited for the job. Those who enjoy solving complex problems and have a passion for AI are most likely to be satisfied in Machine Learning Engineer roles.

What qualities make a good Machine Learning Engineer?

Being a Machine Learning Engineer often works right at the forefront of cutting-edge technology. Because of this, you want to hire someone who is willing to learn and grow with the position. Machine Learning Engineers often work under tight deadlines and in fast-paced environments, so it's important that they're able to work well under pressure. Additionally, they should be willing to learn new skills and adapt to changing technologies. Also, outline all the perks and benefits the position provides.



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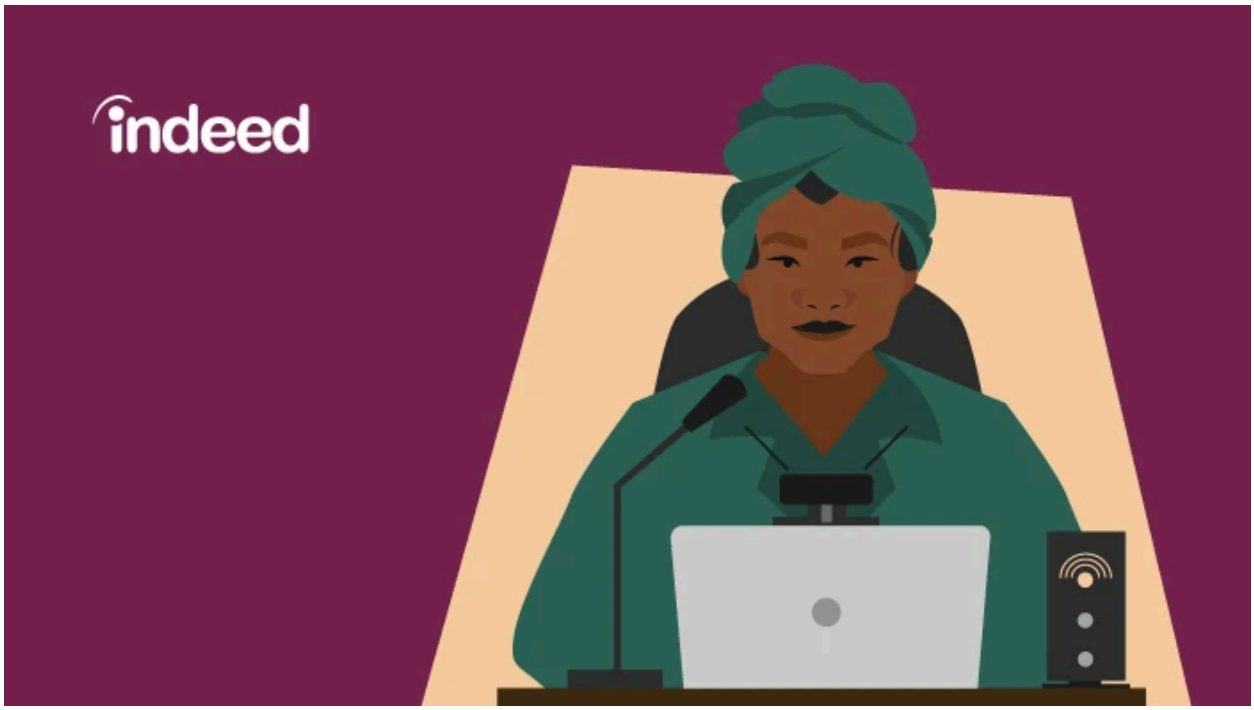
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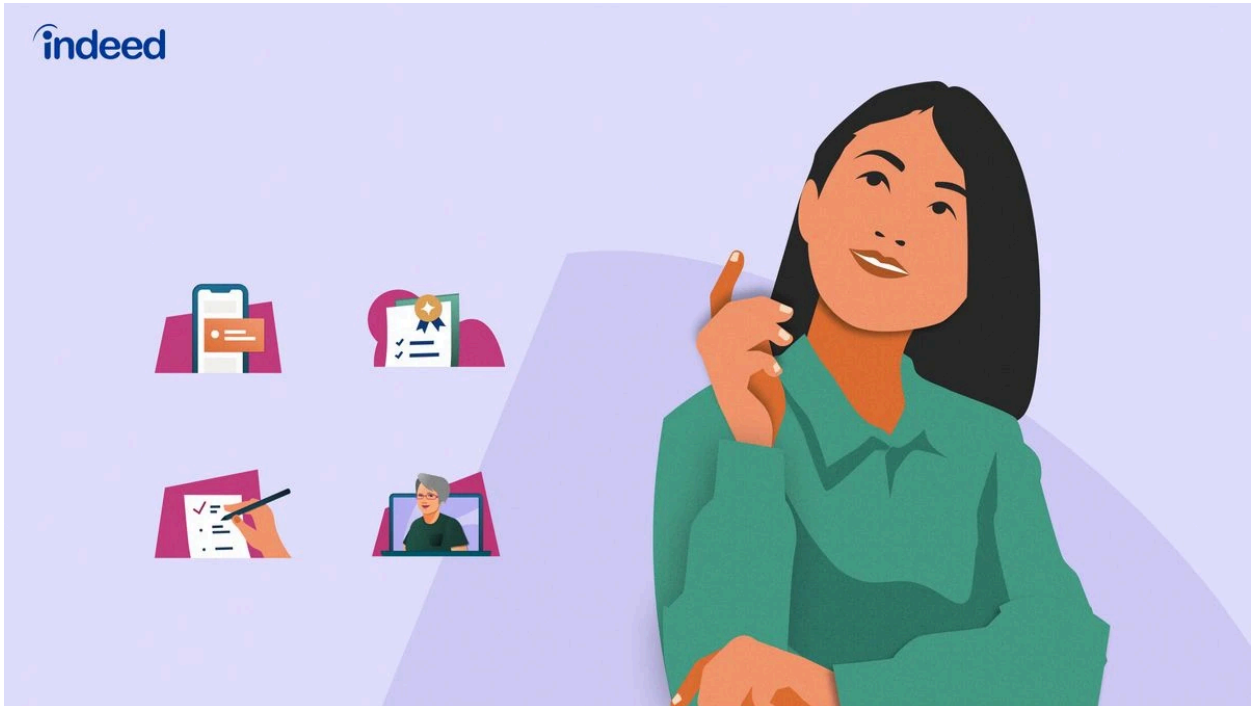
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